

breathbuilt

6 BREATHING TECHNIQUES. THE COMPLETE GUIDE.

TECHNIQUE	PATTERN	DURATION	EFFECT	BEST FOR
● Box Breathing	4s in, 4s hold, 4s out, 4s hold	~5 min	Calming, focusing	Pre-performance, stress
● 4-7-8	4s in, 7s hold, 8s out	~1 min	Sedating	Sleep, acute anxiety
● Physiological Sigh	2s in, 1s sniff, 6s out	~30 sec	Immediate calm	Emergency reset
● Resonant	5.5s in, 5.5s out	5-20 min	Balancing	Daily nervous system training
● Wim Hof	30 breaths + hold + recovery	~6 min	Activating	Morning energy
● Cyclic Sighing	3s in, 1s sniff, 8s out	5 min	Mood boost	Daily practice

This guide is for educational and wellness purposes only. It is not medical advice. Consult your doctor before starting any breathing practice, especially if you have heart, lung, blood pressure, or anxiety conditions.

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● Box Breathing

Focus and calm under pressure

PATTERN

Inhale 4s → Hold 4s → Exhale 4s → Hold 4s · Repeat for ~5 minutes

ORIGIN

Also called square breathing or four-square breathing. Popularized by Mark Divine, a former Navy SEAL commander who documented its use in BUD/S training. The equal-ratio pattern has roots in pranayama traditions, but its modern use in tactical performance training made it mainstream.

HOW IT WORKS

The equal timing across all four phases activates the vagus nerve and shifts the nervous system from sympathetic (fight-or-flight) to parasympathetic (rest-and-digest) dominance. The holds on empty lungs build CO2 tolerance, which trains the brain to stay calm under physiological stress.

BENEFITS

- Lowers heart rate and blood pressure in acute stress
- Improves focus and sustained attention under pressure
- Reduces cortisol (Frontiers in Psychology, 2017)
- Zero equipment, any position, any environment

Tip: Breathe from your belly, not your chest. Start with 2 minutes and build to 5. The hold after exhale is the most important phase.

● 4-7-8 Breathing

The natural tranquilizer for the nervous system

PATTERN

Inhale (nose) 4s → Hold 7s → Exhale (mouth) 8s · 4 cycles max for beginners

ORIGIN

Developed by Dr. Andrew Weil at the University of Arizona Center for Integrative Medicine. Derived from the yogic practice of pranayama. Weil has called it the most powerful anti-anxiety tool he has encountered in decades of practice. The ratio (4:7:8) matters more than the absolute seconds.

HOW IT WORKS

The extended exhale-to-inhale ratio (2:1) increases vagal tone and parasympathetic activation. The 7-second hold forces your lungs to absorb more oxygen. Over time, the practice builds CO2 tolerance and recalibrates your baseline stress response.

BENEFITS

- Powerful sleep aid, many report falling asleep in minutes
- Fastest anxiety reduction from the 2:1 exhale ratio
- Enhanced HRV (Int. Journal of Psychophysiology, 2019)
- Only 4 cycles needed, under 60 seconds total

Tip: If the 7-second hold is too long, scale the ratio down (2-3.5-4) and build up. Inhale through nose, exhale through mouth only. Weil says don't exceed 4 cycles in the first month.

● Physiological Sigh

The fastest voluntary stress reset known to science

PATTERN

Inhale (nose) 2s → Short sniff (nose) 1s → Long exhale (mouth) 6s · 1-3 reps

ORIGIN

A natural reflex discovered in the 1930s. Your body does this involuntarily when you cry, transition into sleep, or hit peak stress. Jack Feldman's lab at UCLA identified the pre-Botzinger complex, the brainstem region generating respiratory rhythm, and the specific neurons within it that produce sighs. Andrew Huberman and the Stanford team brought it into mainstream awareness through their 2023 research.

HOW IT WORKS

The double inhale reinflates collapsed alveoli (tiny air sacs in the lungs), maximizing the surface area available for gas exchange. The long exhale then offloads the maximum amount of CO₂ in a single breath. This is why one rep can shift your entire state. It is a hardware reset, not a software trick.

BENEFITS

- Works in a single breath, the fastest method available
- No learning curve, your body already knows this pattern
- Maximizes CO₂ removal per breath cycle
- Effective in real-time situations, not just as daily practice

Tip: The second inhale is a sniff, not a full breath. Think of it as topping off the last 20% of your lung capacity. The exhale should be longer than both inhales combined.

● Resonant Breathing

Where your heart, lungs, and blood pressure sync up

PATTERN

Inhale (nose) 5.5s → Exhale (nose) 5.5s · No holds · 5-20 minutes

ORIGIN

Luciano Bernardi at the University of Pavia discovered in 2001 that rosary prayer and yoga mantras naturally produce ~6 breaths per minute, a rate that enhances heart rate variability and baroreflex sensitivity. Stephen Elliott developed Coherent Breathing at ~5 breaths per minute. James Nestor synthesized the research in "Breath" (2020), identifying ~5.5 breaths per minute as a convergence point across contemplative traditions worldwide.

HOW IT WORKS

At approximately 5.5 breaths per minute, your cardiovascular, respiratory, and autonomic systems enter a state called coherence. Heart rate variability reaches its peak, blood pressure stabilizes, and baroreflex sensitivity improves. This is a training adaptation that compounds over weeks of consistent practice.

BENEFITS

- Maximizes HRV, the strongest biomarker of stress resilience
- Meditative state without meditation experience
- Lowers blood pressure (Zaccaro et al., 2018)
- Cumulative benefits compound over 4-8 weeks

Tip: Start at 4 seconds in, 4 seconds out and gradually slow to 5.5. Breathe from the diaphragm. This is a long-term practice, real shifts happen at 4-8 weeks of daily use.

● Wim Hof Method

Controlled activation for energy, immunity, and mental toughness

PATTERN

30 deep rapid breaths → Hold on empty lungs (as long as comfortable) → Recovery breath, hold 15s · 3 rounds

ORIGIN

Developed by Wim Hof ("The Iceman"), a Dutch athlete who holds 26 world records for extreme cold exposure. His method combines specific breathing, cold exposure, and meditation. The 2014 Radboud University study (PNAS) was the first to show that trained practitioners could voluntarily influence their innate immune response, something previously considered impossible.

HOW IT WORKS

The 30 rapid breaths create temporary respiratory alkalosis (increased blood pH), which enables extended breath holds. The hold triggers a measurable adrenaline response. The recovery breath re-oxygenates. The combined effect modulates the autonomic nervous system and inflammatory markers. This is an activating technique, not calming.

BENEFITS

- Acute energy boost with measurable adrenaline spike
- Modulates immune and inflammatory response (PNAS, 2014)
- Builds mental toughness and discomfort tolerance
- Compounds with cold exposure for greater adaptation

SAFETY WARNING

Never practice in or near water. Shallow water blackout has caused deaths. Never while driving, standing, or anywhere passing out could cause injury. Sit or lie down only. Do not practice if you have epilepsy, are pregnant, have a history of seizures, or have serious cardiovascular conditions. Consult your doctor first. Tingling and lightheadedness are normal. Stop immediately if you feel pain.

● Cyclic Sighing

Clinically proven to improve mood more than meditation

PATTERN

Inhale (nose) 3s → Short sniff (nose) 1s → Long exhale (mouth) 8s · Repeat for 5 minutes daily

ORIGIN

Validated in a 2023 randomized controlled trial published in Cell Reports Medicine (Balban, Neri, Kogon, Weed, Huberman, et al., Stanford). 108 participants practiced 5 minutes daily for 28 days across four groups: cyclic sighing, box breathing, cyclic hyperventilation, and mindfulness meditation. Cyclic sighing produced the greatest improvement in positive affect and the greatest reduction in resting respiratory rate.

HOW IT WORKS

Structurally identical to the physiological sigh, but repeated continuously as a daily practice. The extended exhale (roughly 2x the inhale duration) activates the parasympathetic system on every cycle. The researchers noted that breathwork's advantage over meditation may come from the active control over a physiological process, creating a sense of agency rather than passive observation.

BENEFITS

- Clinically shown to improve mood more than meditation
- Reduces resting respiratory rate over time
- Only 5 min/day for measurable results
- Best "default" technique if you only pick one

Tip: Do not confuse with the single physiological sigh. One sigh is for emergencies. Cyclic sighing is for daily practice. Consistency at 5 min/day beats longer sessions done sporadically.

Safety and When to Use What

GENERAL GUIDELINES

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IF BREATHING INCREASES ANXIETY

For some people, focusing on the breath can increase anxiety rather than reduce it. This is normal and does not mean something is wrong. If counted breathing feels uncomfortable, try the physiological sigh instead. One breath, no counting, no timing. If breathwork consistently increases distress, consult a mental health professional.

QUICK DECISION GUIDE

SITUATION	BEST TECHNIQUE	WHY
Panic, bad news, heart pounding	Physiological Sigh	One breath, instant effect, no setup
Can't fall asleep, racing mind	4-7-8	Under 60 seconds, sedating effect
Before a meeting, interview, stage	Box Breathing	5 minutes, sharpens focus under pressure
Daily mood and resilience	Cyclic Sighing	5 min/day, strongest evidence for mood
Long-term nervous system training	Resonant	Maximizes HRV, compounds over weeks
Morning energy, pre-workout	Wim Hof	Activating, adrenaline response

PAIRING SUGGESTIONS

Morning + Night

Wim Hof in the morning for energy. 4-7-8 at night for sleep. Opposite ends of the spectrum, complementary effects.

Emergency + Daily

Physiological sigh for acute moments throughout the day. Cyclic sighing as a consistent 5-minute daily practice. Same underlying mechanic, different applications.

Starter Stack

If you are new to breathwork, start with these three: Physiological Sigh (instant use, no training), Box Breathing (foundational, builds discipline), and Cyclic Sighing (daily practice, best evidence). Add Resonant and Wim Hof once these feel natural.

